

ENHANCED FORECASTING OF FLOOD EVENTS BY INTEGRATING ARIMA IN TIME SERIES DECOMPOSITION: AN APPLICATION AT THE DELAWARE RIVER, TRENTON, NJ

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Abstract

Flood forecasting is crucial to saving billions of dollars by fostering the resilience of a society and saving countless lives. Heavy rainfall and warmer temperatures escalate the amount of water discharge and the probability of ice jams in the Delaware River, this intensifies Trenton, New Jersey's risk of flooding. Although it is difficult to predict short-term outcomes, this research utilized time series analysis and multiple autoregressive integrated moving average (ARIMA) for flood predictions in Trenton, New Jersey. Time series decomposition has been applied to separate the noise from the signal of available hydrogeological and climatic data from the time span of April of 2005 to August of 2018. Data includes precipitation, wind speed, wind direction, air temperature, water temperature, tides, depth to groundwater, and water discharge. Time series were separated into different time scale components such as the long-term, seasonal-term, and short-term utilizing the Kolmogorov-Zurbenko filter. Lag operators were applied to the time series decomposition using a data mining software, KNIME, to maximize the performance model of both hydrogeological and climatic fluctuations. ARIMA was implemented to both the raw data and the difficult to predict short-term component to enhance the overall model and strengthen the association between water discharge and hydrogeological data as well as climatic data. The raw data modeling with a multiple linear regression model (MLR) provides a total explanation of 54%, while the application of the Time Series Regression (TSR) model and the ARIMA model in the decomposed data provides a total explanation up to 87%. This methodology can be applied for the prediction of the water discharge at multiple sites.

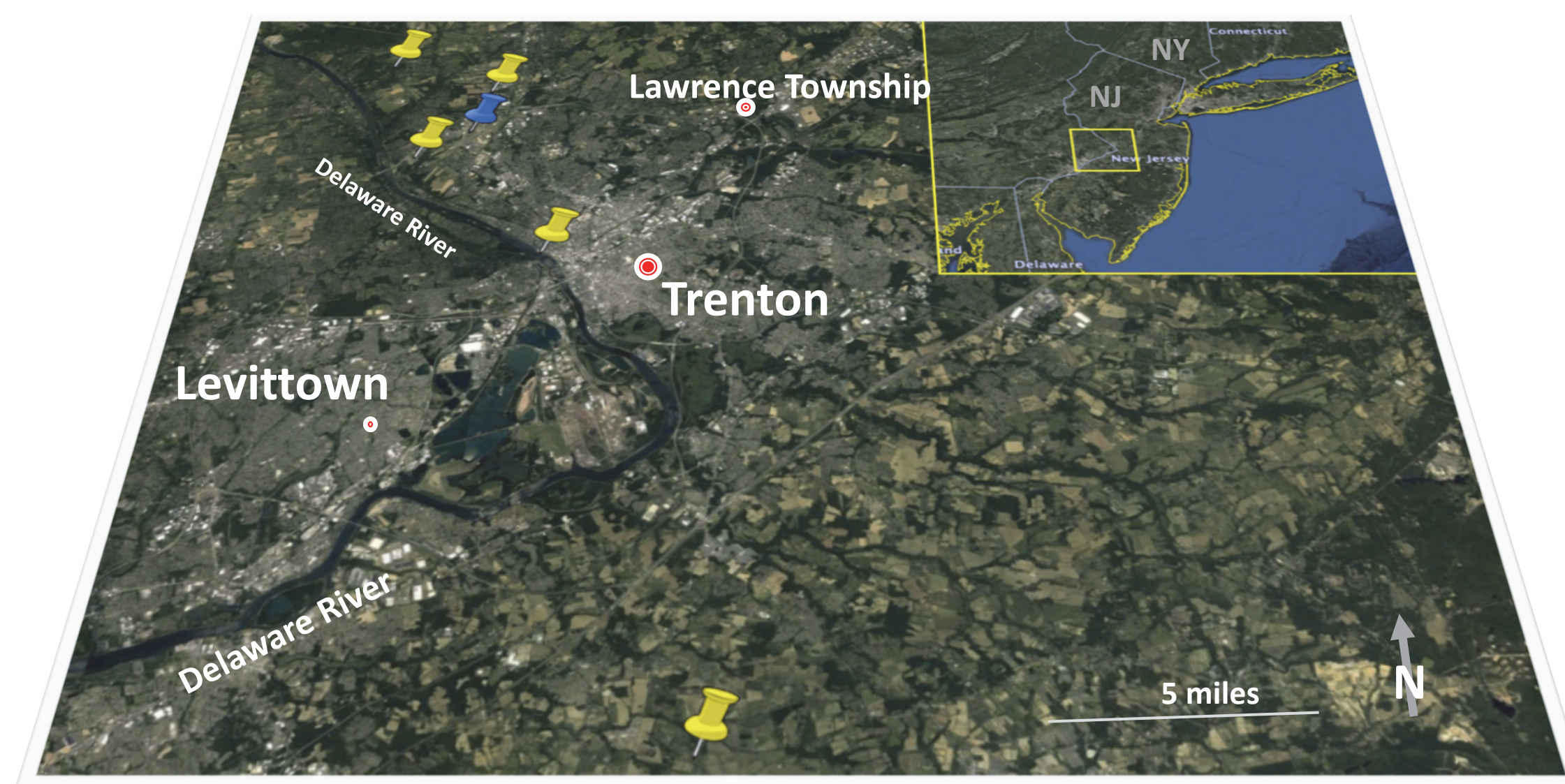


Figure 1: Aerial image obtained from Google Earth of the study area and the utilized monitoring sites USGA data (yellow) and NOAA data (blue).

Introduction

Natural hazards such as flood events may severely impact the socio-economic structure [1], and in New York along the Mohawk River and in New Jersey at the Delaware River and other tributaries, flooding and related prediction efforts of those flood events have been made [2]. Flood forecasting research will enhance a resilient society in highly occupied regions such as in New Jersey State. We aim to separate individual patterns and decrease interferences originating from covariance structures between the components of the time series. We compare three different statistical models for prediction of flood such as the MLR, TSR, and ARIMA to facilitate a flood warning system for several sections at the Delaware River in New Jersey (Figure 1).

Methods

- Data were retrieved and pre-processed using KNIME.
- Atmospheric and hydrogeological variables, such as water discharge, groundwater precipitation and temperature.
- Decomposition of time series and separation of the different signals with minimum interferences between the components of the time series [3,4].
- Raw data were decomposed to a long-term trend (fluctuations longer than a given threshold), seasonal-term (intra-annual fluctuations), and a short-term component (short-term variations) [5, 6].
- Multiple Linear Regression (MLR), Time Series Regression (TSR), and ARIMA was performed.

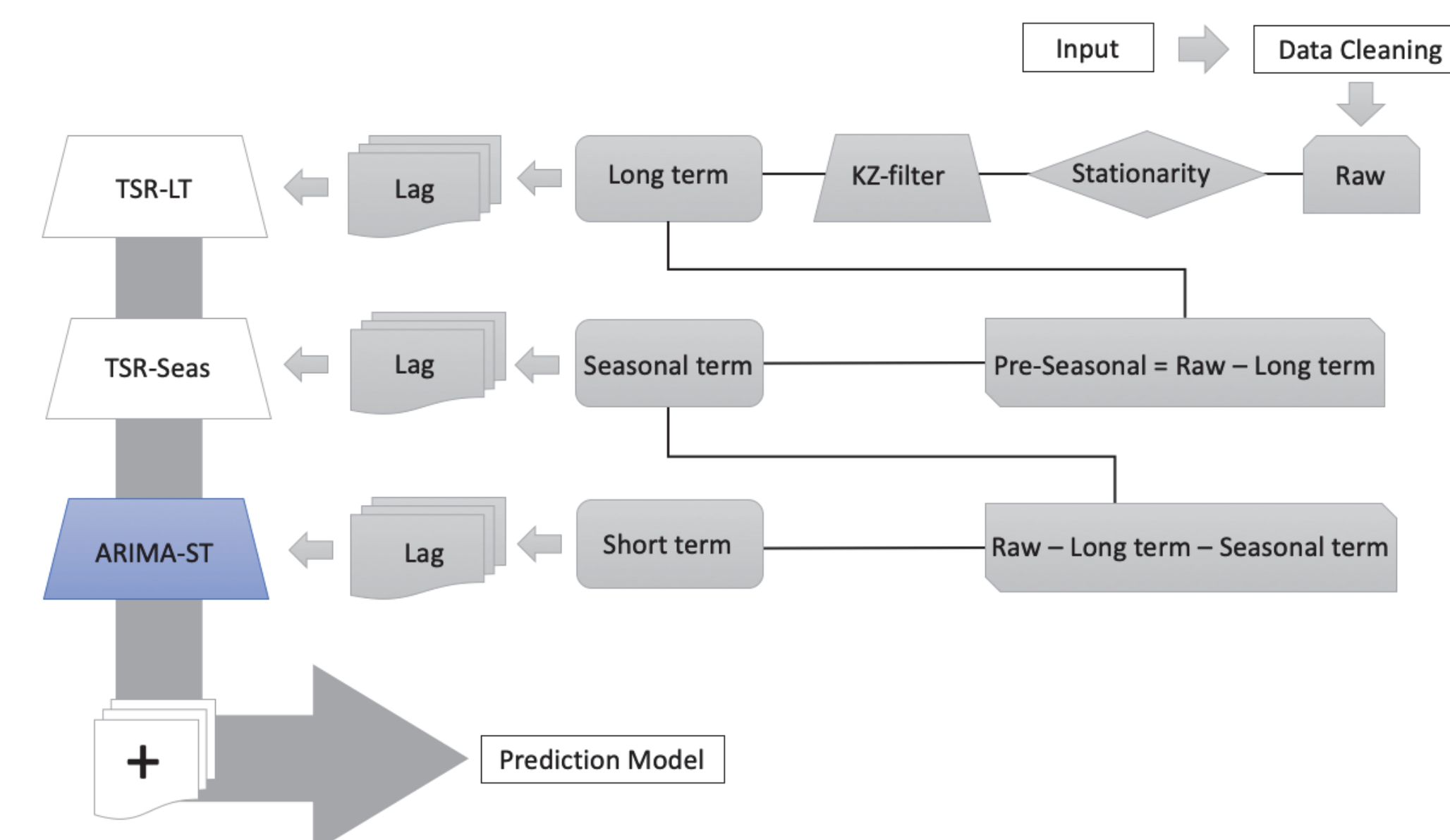


Figure 2: A flowchart displaying the methodology which includes the decomposition of time series utilizing the Kolmogorov-Zurbenko filter. Lag operators and time series regression (TSR) were applied to the long term, and the seasonal term, while autoregressive integrated moving average (ARIMA) was applied to the short term.

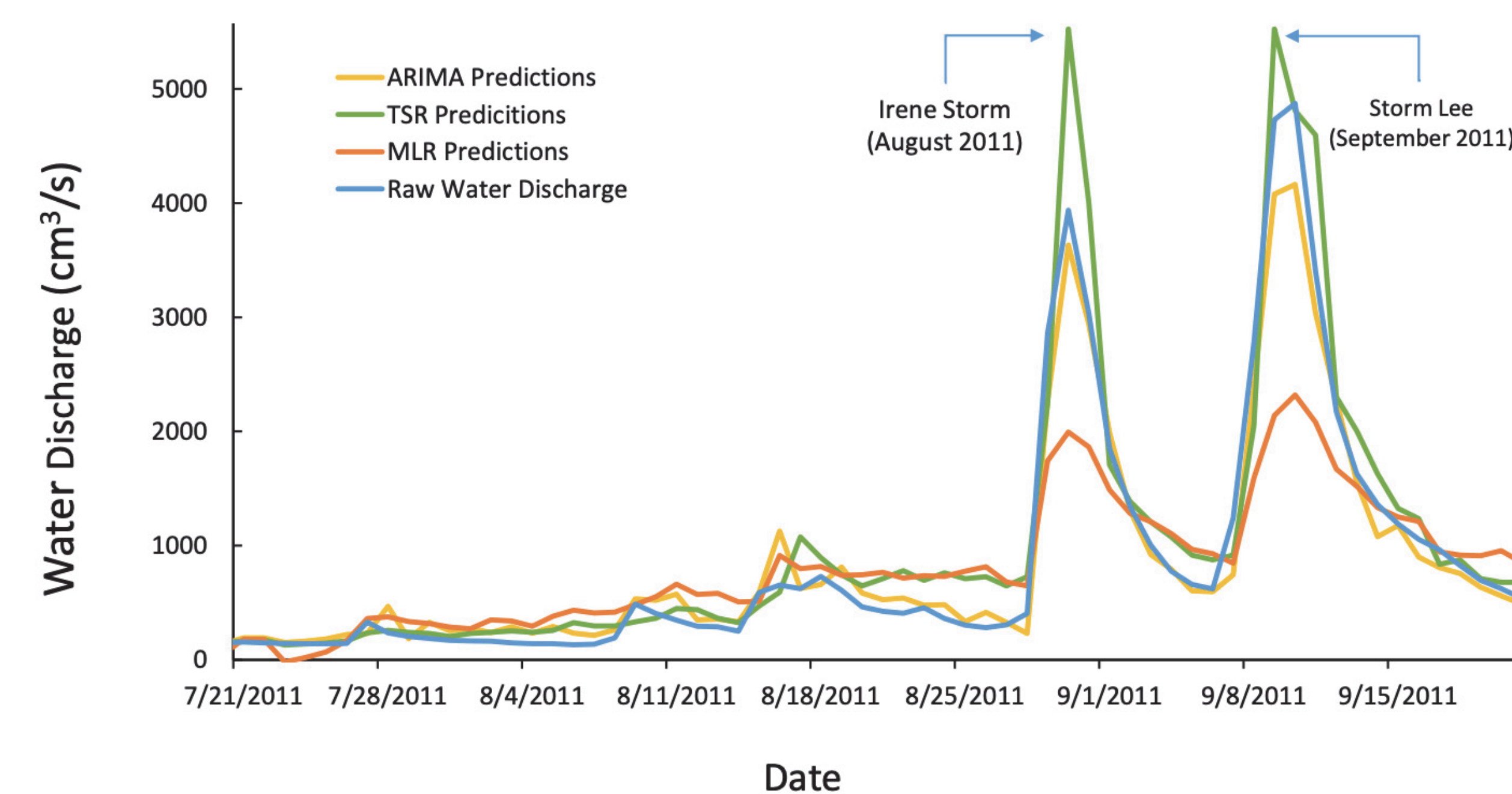


Figure 3: Various prediction models of the raw (real) water discharge data (blue line) derived from the ARIMA model (yellow line), the TSR model (green line), and the MLR model (orange line). The peak in water discharge in August resulted by the Hurricane Irene, while the peak in September is a result of Tropical Storm Lee.

References

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Results

	Raw Data	LT	Se	ST	ST ARIMA
Coefficient Determination (R2)	0.54	0.87	0.46	0.44	0.93
Variance %		65.22	3.26	31.52	31.52
Explanation %		56.55	1.51	13.99	29.38
Total Explanation % (TSR)					72.05
Total Explanation % (TSR/ARIMA)					87.44

Table 1: (Above) Summary of the explanation in the water discharge time series.

Table 2: (Right) Summary of the coefficient determination (R²) in the water discharge time series for the various time scale components.

	LT	Se	ST
R ² (MLR)	0.74	0.50	0.29
R ² (TSR)	0.93	0.58	0.41

Figure 4: (Right) The predicted hybrid ARIMA (black line) with the decomposition data plotted against the raw values (grey line) of the water discharge time series of the long-term component for USGS station 400242074422301.

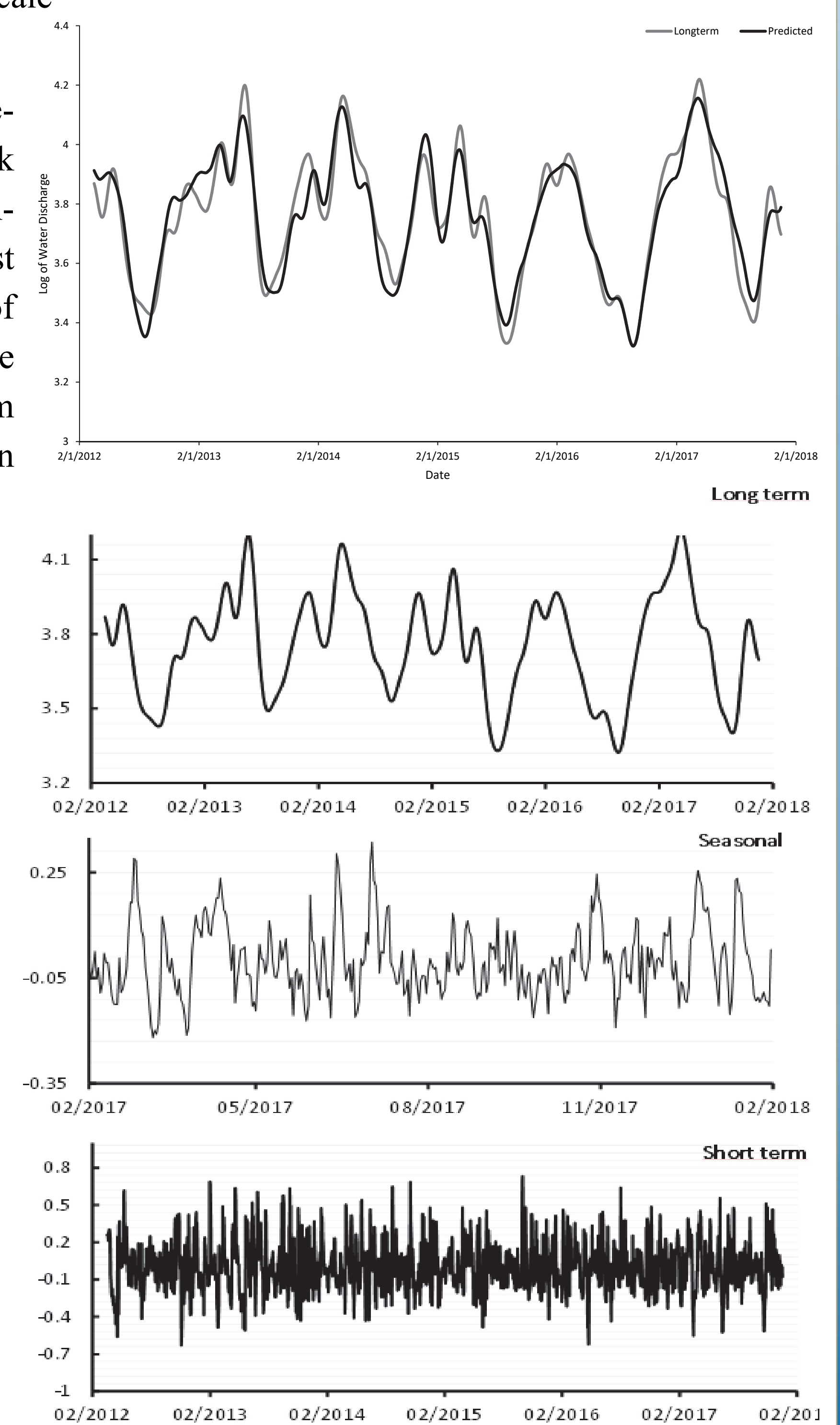


Figure 5: (Right) Raw water discharge data decomposed into long-term, seasonal-term and short-term.

Discussion

- Floods can be observed using laboratory stimulation techniques without requiring a visit the flood site.
- An evaluation of flood events in this region can be assessed by looking at various atmospheric and hydrogeological variables.
- Time series decomposition separates variables into long-term, seasonal-term, and short-term to minimize the amount of noise within the prediction model.

Conclusion

Time series decomposition is a powerful statistical technique utilized for flood prediction; however, the model may improve with the integration of an autoregressive integrated moving average (ARIMA).